BTS–I(S)/II(S)–11.19–1437 Reg. No.

C

B.Tech. Degree I Semester/II Semester Supplementary Examination November 2019

ME/SE GE 15-1203 A ENGINEERING GRAPHICS

(2015 Scheme)

Time: 3 Hours Maximum Marks : 60

(5 × 12 = 60)

I. The area of a field is 50,000 sq m. The length and the breadth of the field, on the map is 10 cm and 8 cm respectively. Construct a diagonal scale to show single metre and mark the length of 235 metre. What is the R.F. of the scale?

OR

II. Construct a parabola in parallelogram of side 100 mm × 45 mm and with an included angle 75°. Find the focus and directrix.

III. A line PQ has its end P, 10 mm above the HP and 20 mm in front of the VP. The end Q is 35 mm in front of the VP and the front view of the line measures 75 mm. The distance between the end projectors is 50 mm. Draw the projections of the line and find its true length and the true inclinations with VP and HP.

OR

IV. Draw the projections of a regular hexagon of 25 mm side, having one of its side in the HP and inclined at 60° to VP, and its surface making an angle 45° with the HP.

V. Draw the projections of a tetrahedron of edges 50 mm when its rest on one of its edges on VP with the resting edge normal to HP. One of the faces containing the resting edge makes an angle 30° with the VP.

OR

VI. A hexagonal pyramid of base side 25 mm and axis 60 mm rests on its base on the HP with two base edges perpendicular to the VP. It is cut by a plane perpendicular to the VP and inclined at 30° to the HP meeting the axis at 25 mm from the vertex. Draw the elevation, sectional plan and the true shape of the section.

VII. A cone of base diameter 100 mm and height 130 mm rests with its base on HP. It is cut by a section plane inclined at 30° to HP and perpendicular to VP. Draw the development of the truncated cone if the section plane bisects the axis of the cone.

OR

VIII. A vertical cylinder of 80 mm diameter is completely penetrated by another cylinder of 60 mm diameter, their axes bisecting each other at right angles. Draw their projections showing curves of penetration, assuming the axis of the penetrating cylinder to be parallel to the VP.

IX Draw the isometric view of a pentagonal pyramid, side of base 40 mm and height 80 mm which rests with base centrally on a cylinder of diameter 120 mm and height 40 mm.

OR

X. Draw the perspective projection of a pentagonal prism of side 25 mm and length 50 mm, lying on one of its rectangular faces on the ground plane and one pentagonal face touching the picture plane. The station point is 55 mm in front of the picture plane and lies in the central plane which is 75 mm to the left of the centre of the prism. Station point is 30 mm above the ground plane.

***